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Paleomagnetic study of Cretaceous rocks from the Yangtze block, central Yunnan, China: implications for the India–Asia collision

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ABSTRACT

More than 200 samples for this study were collected at 30 sites from the Cretaceous Matoushan and Puchanghe Formations around the city of Chuxiong (25°N,101.5°E) Twenty-four sites have characteristic directions with a high temperature component above 350°C Paleomagnetic directions from 21 sites of the Cretaceous formations reveal clockwise deflection in declination ($D = 44.6°$, $I = 41.3°$, and $\alpha_{95} = 10\ 7°$). The reliability of the direction is ascertained through a positive fold test, and the presence of normal and reversed polarities. Comparison with paleomagnetic data from other areas indicates that the Chuxiong area was rotated clockwise through $26° \pm 17°$ with respect to the eastern part of the Yangtze block, whereas it was subjected to more than 44° of clockwise rotation with respect to the neighboring blocks of Lhasa ($56° \pm 21°$) and western Sichuan ($44° \pm 16°$) since Cretaceous time. The clockwise rotation of Chuxiong is a reflection of rotational motion of the large region including the southeastern Qiangtang Terrane and the Jinsha suture–Red River fault. Paleomagnetic data and tectonic features suggest that the rotation occurred prior to the folding in the central Yunnan province, probably during the Paleogene This large rotation is a spectacular example of the deformation in the southeastern part of the Asian continent due to the collision of India.

1. Introduction

The collision of the continents of India and Asia was one of the greatest Neogene tectonic events. The Indian continent moved northward since splitting from Antarctica and collided with the Asian continent at about 45 Ma [1–5]. The collision appears to have given rise to crustal deformation in the Asian continent, and formed the geomorphologically unusual shape of the Tibetan plateau and its vicinity in front of the Indian continent. The deformation caused by the collision of India has been studied by both laboratory and numerical experiments, in which the rigid die of India penetrates into the continuum of Asia [6–11]. There have also been many geological and geophysical field observations of the deformation in the Tibet region [e.g. [12–14]. Elucidating aspects of deformation are important for understanding the dynamic characteristics of non-plate-like bodies of continental lithosphere and for describing the tectonic history of Asia.

Paleomagnetic studies have also played an important role in evaluating deformation in continents. Major deformation features in the Tibetan plateau and its vicinity are divided into two categories on the basis of the paleomagnetic data [15–21]—shortening of the continent in front of

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the indenting continent, and deformation on its flanks. Study of the former has been progressing since the beginning of surveying on the Tibetan plateau [15–18], whereas work on the latter has been undertaken only more recently [19–21]: tectonic deformation is now known from the Markam area, in the Three Rivers region on the northeastern side of the indented block [19,21].

We have focused on the deformation in the central part of the Yunnan (Fig. 1). This area is south of the Three Rivers region and corresponds to the deformed area of the eastern flank of the indenting continent that is expected from the experiments [e.g. 10]. We undertook an expedition to the Chuxiong area in central Yunnan in 1989, during which more than 200 samples were

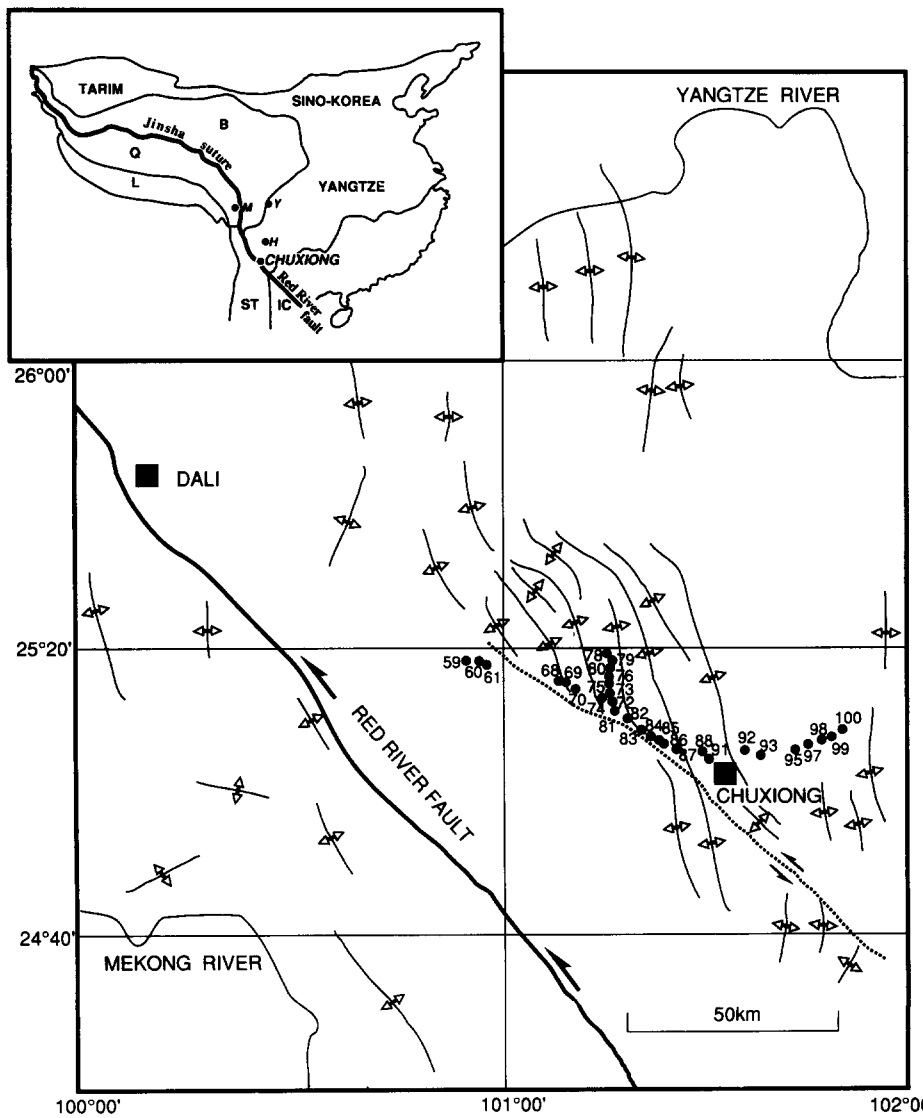


Fig. 1. Regional map of central Yunnan (redrawn from [28]) around Chuxiong, with sampling sites (dots) indicated. Bold line indicates the Red River fault. Dotted line indicates a minor fault through the Chuxiong area. Tertiary anticlines are also shown. A sketch map of the tectonic blocks of the southeastern part of the Asian continent is indicated in the inset. The bold line indicates the Jinsha suture–Red River fault. *B* = Bayan Kala Terrane; *Q* = Qiangtang Terrane; *L* = Lhasa Terrane, *ST* = Shan-Thai block; *IC* = Indochina block; *M* = Markam; *Y* = Ya'an; *H* = Huili. Ya'an and Huili are located in western Sichuan.

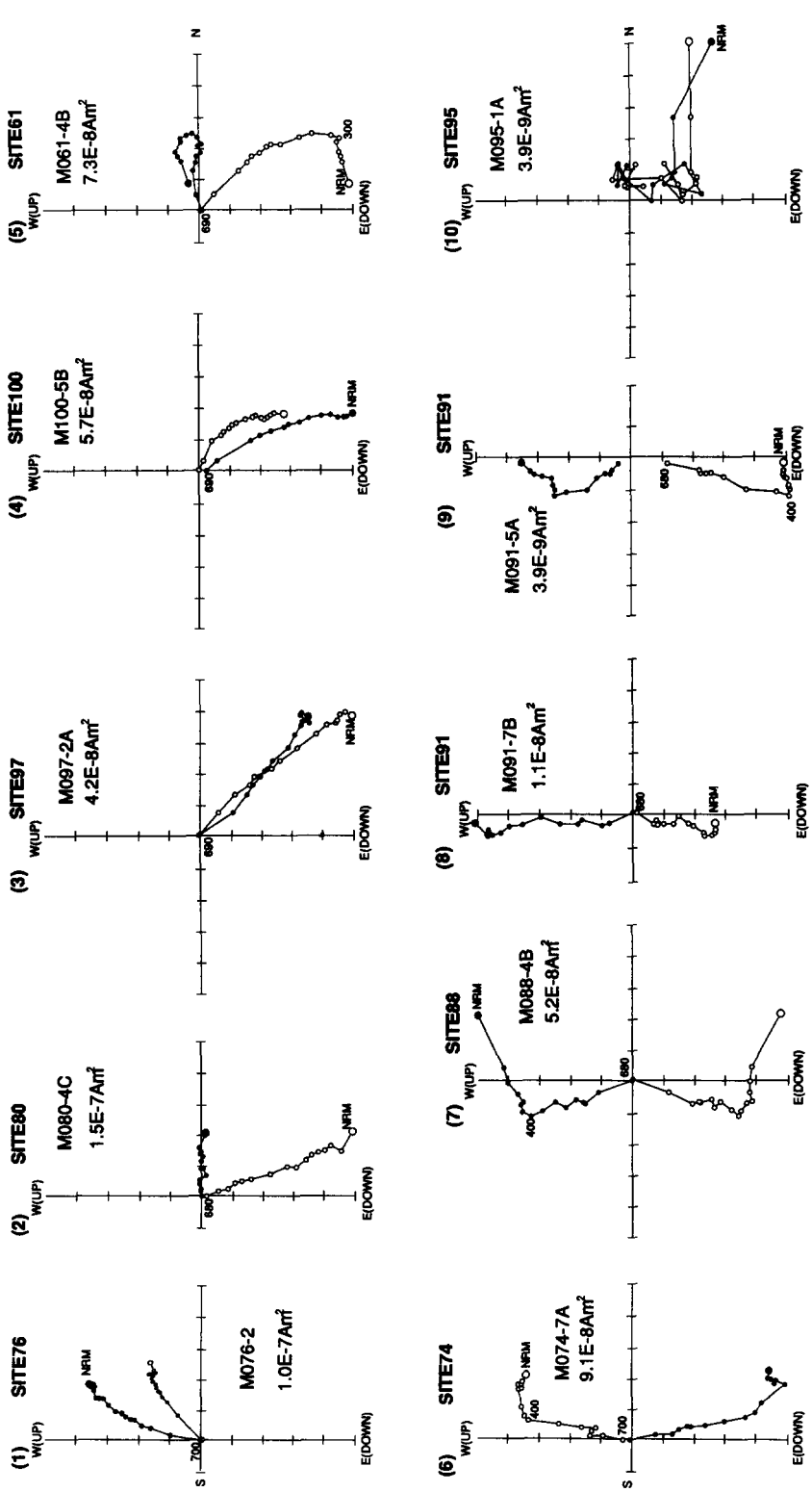


Fig. 2. Orthogonal projections of magnetization vector end points during thermal demagnetization experiments for specimens in three categories (A, B and C) (see text). (1–4) = Category A; (5–7) = Category B, (8–10) = Category C. Open symbols show the magnetic vector projection on to the vertical plane, solid symbols on to the horizontal plane. Numbers beside symbols represent temperature (°C). Tectonic tilt correction has not been applied.

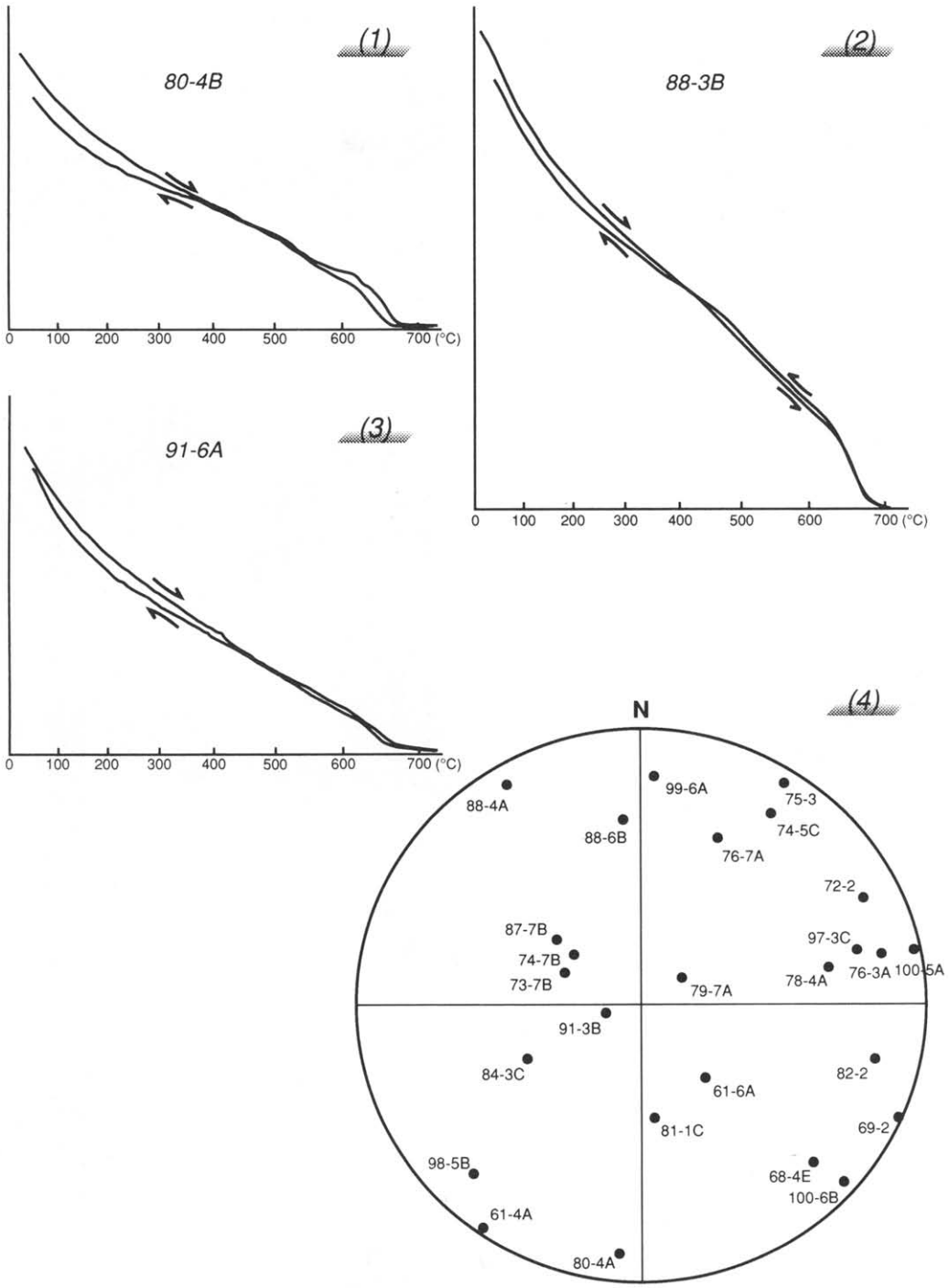


Fig. 3. Thermomagnetic analysis performed in air, and the anisotropy of the magnetic susceptibility (1–3) Thermomagnetic curves ($Js-T$) of specimens 80–4B, 88–3B and 91–6A, which represent those of categories A, B, and C (see text), respectively. High field saturation magnetization (Js) is conventionally in arbitrary units. The arrows indicate heating and cooling. Thermal decay of saturation isothermal remanent magnetization (Js) show that magnetization is not completely removed by 650°C, indicating the presence of hematite. (4) Orientation of maximum susceptibility axes after tectonic tilt corrections. All the directions are plotted on the lower hemisphere. Numbers besides the symbols are the names of sites specimens (e.g , 61–6A = site–specimen). Equal-area projection.

collected from Cretaceous red beds in an attempt to detect intracontinental deformation paleomagnetically.

2. Geological setting and sampling

The Chuxiong area is located in central Yunnan, which forms the southwestern tip of the Yangtze block (Fig. 1). This block is bounded to the west by the Shan-Thai block and to the southeast by the Indochina block [22,23]. The Shan-Thai and Indochina blocks formed the stable southern margin of Asia after suturing with the Yangtze block, which took place as early as the Triassic [24,25]. The boundary between the Yangtze block and the other two blocks is the

TABLE 1

Paleomagnetic results for the Cretaceous rocks of Chuxiong

Site		Demag component (°C)		N	In-situ				Tilt-corrected				Locality	
					D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)	Lat (°N)	Long. (°E)
Matoushan Formation (Upper Cretaceous)														
72	A	HT	0–680	5	−82.7	46 7	193.4	5 5	50.2	44.3	193.4	5.5	25.20	101 26
73	A	HT	100–680	5	74 6	26.0	183 9	5.7	23.6	51 0	183.9	5.7	25.21	101.27
74	B	HT	300–700	5	69 1	−26.0	35 8	13.0	67.3	31.5	35.8	13 0	25.23	101.27
		LT	200–450	2	−1.7	−33 3	–	–	36 9	−18.9	–	–		
75	A	HT	150–700	5	72 2	19.6	56 2	10 3	37.4	62 9	56.2	10.3	25.24	101 26
76	A	HT	350–700	5	−71 6	−6.1	58 0	10 1	−64.4	−40 8	58.0	10.1	25.28	101 27
78	A	HT	100–700	5	37.6	10 2	52 0	10 7	24 3	32.9	52.0	10.7	25.36	101 24
79	A	HT	100–700	5	35 7	54 6	203 7	5 4	12 2	49 0	203.7	5.4	25.30	101.28
80	A	HT	100–680	5	8 6	65 5	80.2	8 6	37.2	46.7	80.2	8.6	25.30	101 28
93*	A	HT	0–680	5	45 8	76 6	406.1	3.8	−102 7	43 7	406 1	3.8	25 05	101.68
99	A	HT	250–700	5	27 7	21 8	55.8	10.3	30.5	41.3	55 8	10.3	25.14	101.87
100	A	HT	0–690	5	72.2	22 4	94.9	7.9	58 4	31 5	94.9	7.9	25 15	101.89
mean		HT		10	56 8	31.7	3.3	31.6	45 6	46.6	12.4	14.3	25 00	101.50
Puchanghe Formation (Lower Cretaceous)														
59	C												25.32	100.94
60	C												25.30	100 95
61	B	HT	150–690	5	−7.4	53 0	90.0	8.1	18.2	25.3	90.0	8.1	25.30	100.96
		LT	0–400	2	−63 2	−4 3	–	–	128 3	42 9	–	–		
68	A	HT	200–680	5	−39.8	38 2	22.1	16.7	−2 4	25 2	22.1	16.7	25.25	101.16
69	A	HT	0–700	5	18 7	60.5	20.6	17.3	30 6	38 5	20.6	17.3	25.23	101.18
70	C												25.21	101.20
81	A	HT	200–690	5	39 0	42.6	33 9	13.3	25 4	−12.1	33.9	13.3	25.17	101.32
82	A	HT	200–690	5	−114 0	−14.9	115.6	7 2	−108 6	−37 6	115.6	7.2	25.17	101.33
83*	C	HT	200–690	5	−66.1	22.0	182 3	5 7	−40 0	17 2	182.3	5.7	25.14	101.36
84	A	HT	100–700	5	−114.6	−10 9	205 8	5 4	−100.3	−46 9	205.8	5.4	25.13	101.38
85	C												25.15	101.35
86*	A	HT	100–700	5	9 2	3 3	32 3	13.7	−36.8	50.3	32.3	13.7	25 13	101 39
87	A	HT	0–680	5	−10 4	45.4	416 4	3 8	87.2	57 7	416.4	3 8	25.10	101 43
88	B	HT	350–700	5	−105.6	52.9	95 0	7 9	−106.4	−32 8	95.0	7.9	25.08	101 49
		LT	0–450	3	−20.9	24 3	34.6	21.3	−54.4	8 8	34.8	21.3		
91	C	HT	250–680	5	−97.3	52 1	18.3	18.4	28.9	38.7	18.3	18 4	25.05	101 50
		LT	0–400	2	−63.2	−4 3	–	–	−45 5	24.4	–	–		
92	C												25 05	101 68
95	C												25 03	101 74
97	A	HT	0–690	5	32.9	42.5	88 0	8 2	52 5	36.6	88 0	8 2	25 04	101 75
98	A	HT	150–690	5	51.8	−27.4	70 3	9.2	47 5	35.4	70.3	9.2	25.12	101.85
mean		HT		11	28.9	37.1	2.6	35 7	43.7	36.2	8.1	17.1	25.00	101.50
Mean of Matoushan and Puchanghe Formation														
		HT		21	43 4	35 2	2 9	22 9	44 6	41.3	9.8	10.7	25.00	101.50

Red River fault, which connects with the Jinsha suture in Tibet. A bend in the block boundary is quite clear at about 23°30′N, 102°00′E: the strike of the Jinsha suture–Red River fault is N10°W–N35°W to the north of this point, whereas the strike of the Red River fault is N70°W to the south.

Mesozoic rocks occur over a wide area around Chuxiong and Puchanghe in central Yunnan [26,27] and are mainly composed of purple sandstones and siltstones. The Jurassic formations conformably underlie the Lower Cretaceous formations, which are followed disconformably by the Upper Cretaceous formations. The Cretaceous system is divided into four successions, each well dated by fossils. In ascending order these successions are the Gaofengsi, Puchanghe, Matoushan and Jiangdihe Formations. Folding took place in the Paleogene [28], resulting in north–south trending fold axes in the Mesozoic of Yunnan Province.

We collected samples at eleven sites from the Upper Cretaceous Matoushan Formation and at nineteen sites from the Lower Cretaceous Puchanghe Formation (Fig. 1). The Cretaceous formations in which these sampling sites occur consist of massive red sandstones and red mudstones. No internal deformation was observed in these rocks. The bedding attitude at the sampling sites varies from 19° to almost vertical. Hand samples were oriented with a magnetic compass. Each site typically comprised seven independently oriented samples distributed over a distance ranging up to 50 m. The present declination value at each sampling point was evaluated from calculation using the International Geomagnetic Reference Frame of 1989 [29].

3. Paleomagnetism

3.1. Paleomagnetic measurements

In the laboratory, individual specimens, 25 mm in diameter and 25 mm long, were prepared from five samples from each site. Natural remanent magnetization (NRM) was measured with an ScT cryogenic magnetometer. All specimens were subjected to stepwise thermal demagnetization up to 710°C (Fig. 2). The anisotropy of magnetic susceptibility (AMS) was measured on a spinner magnetometer (Schonstedt SSM-1A). The thermomagnetic analyses were carried out with a recording thermomagnetic balance in air in a magnetic field of 0.85 T and with a vibrating sample magnetometer in a vacuum in a magnetic field of 1.0 T (Fig. 3).

Zijderveld component plots [30] and equal-area projections were used to assess the directional stability of each specimen. Principal component analysis [31] was used to estimate the direction presented by the colinear array of five or more data points defining the component of magnetization.

3.2. Demagnetization

Progressive thermal demagnetization of the samples separated the sampling sites into three categories (Fig. 2, Table 1): (A) single component magnetization with high unblocking temperatures, (B) two-component magnetization, and (C) very variable behavior within one site.

Category A All specimens at nineteen sites exhibited single component magnetization (Fig. 2, (1)–(4); Table 1) and their stable NRM directions clustered. Unblocking temperatures ranged around 700°C. The thermomagnetic (Js–T) curves showed that samples of this category had single high Curie temperatures of around 680°C (Fig. 3, (1)). These suggest that the high temperature component magnetization is carried by hematite. Some samples had laboratory-induced remanent magnetization, which was destroyed by 100°C.

Category B Three sites exhibited two distinct magnetization components (Fig. 2, (5)–(7)). Unblocking temperatures of the high temperature component ranged between 650°C and 700°C, indicating that this component is carried by hematite, as found in the single component magnetization. This inference was ascertained by the thermomagnetic analyses (Fig. 3, (2)). The low temperature components were removed in the initial stage of demagnetization by 350°C. A few samples had directions of the low temperature component retrieved by the principal component analysis. The mean direction of the low temperature component of site 88 fell near the present axial dipole field before tilt correction (Table 1). The low temperature component eliminated by 350°C is probably a viscous partial thermorema-

nent magnetization of hematite acquired in the recent geomagnetic field [32,33].

Category C Sites 83 and 91 consisted of samples with single and two component magnetization (Fig. 2, (8)–(9)). Both single component and high temperature components in the two compo-

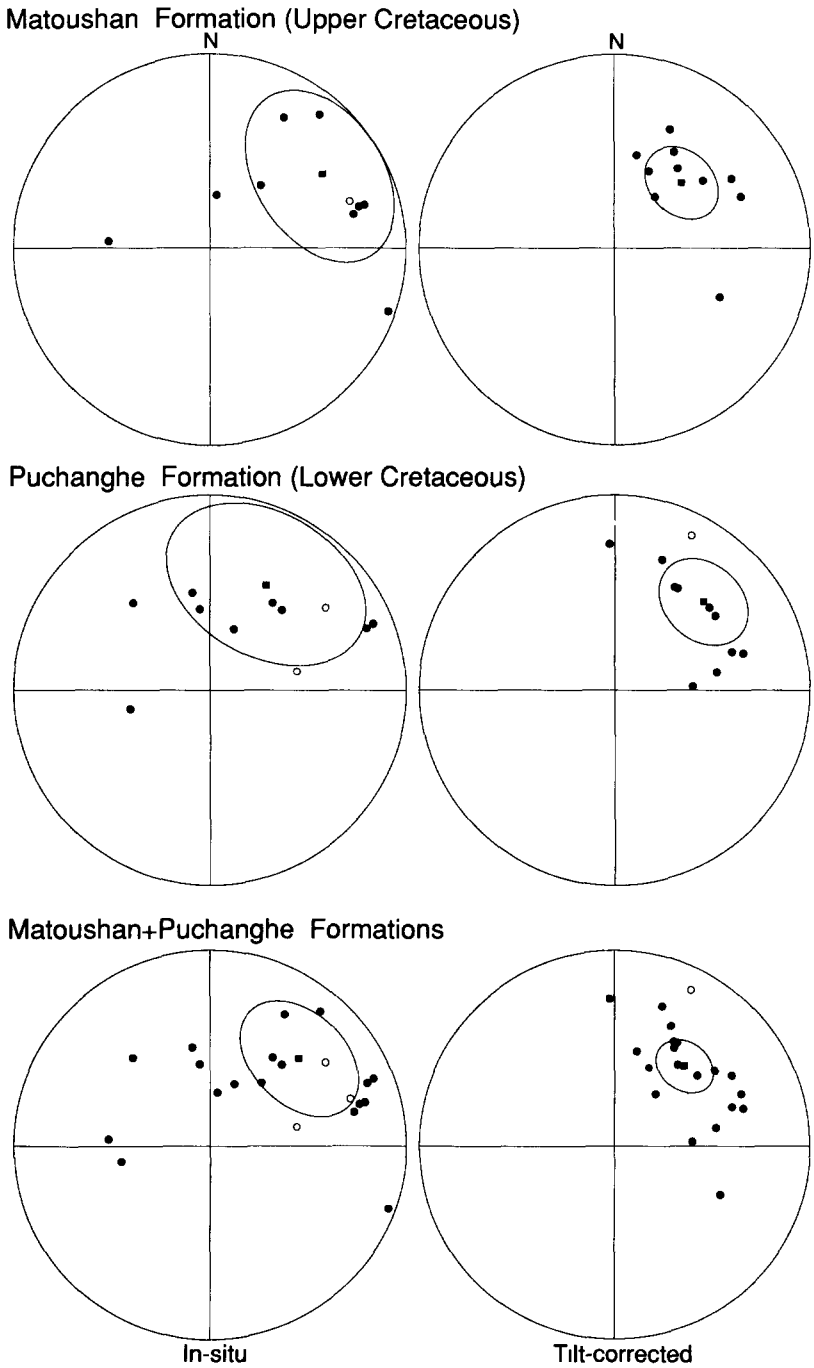


Fig. 4. Equal-area projections of the site mean directions for the Matoushan Formation, the Puchanghe Formation and both formations combined, before and after tilt correction. All the reversed directions have been inverted through the origin. The three data denoted by the asterisk in Table 1 are not shown in this figure. The solid symbols refer to the lower hemisphere, the open symbols to the upper hemisphere. Mean direction for each group is drawn as a square with a circle of 95% confidence.

nent magnetization have similar unblocking temperatures, at around 700°C, and their directions are fairly clustered. Thermomagnetic analyses suggest that the high temperature component is carried by hematite (Fig. 3, (3)).

Two sites (site 85 and 92) had samples with variable component magnetization within one site (single, two and three component magnetization). In most of the samples, none of the paleomagnetic directions of any of the components were identified by principal component analysis because of the fairly erratic demagnetization paths of the paleomagnetic directions. Antipodal directions were revealed in the demagnetization paths of some samples (northeasterly declination with positive inclination and southwesterly direction with negative inclination). These samples would involve both normal and reversed polarity components of the magnetizations, even within a single specimen.

Samples from four sites (site 59, 60, 70 and 95) had weak magnetization (e.g. Fig. 2, (10)), and revealed completely erratic demagnetization paths after their intensity decreased to less than 10^{-4} A/m. No characteristic direction could be isolated from these sites.

3.3. *Characteristic direction*

The characteristic direction could be isolated at 24 sites (Fig. 4, Table 1). We arbitrarily discarded results from any site for which the radius of the 95% confidence circle exceeded 22°.

Matoushan Formation (Upper Cretaceous) Eleven sites have characteristic directions. These directions, with the exception of one from site 93, converge to the northeast ($D = 45.6°$, $I = 46.6°$, $\alpha_{95} = 14.3°$) after tilt correction. We recognize here that the direction of site 76 is a reversed polarity because of its negative inclination. The ten sites (i.e. excluding site 93) provide a positive fold test which is significant at the 99% probability level: the precision parameter k increased from 3.3 to 12.4 ($k_2/k_1 = 3.8$) [34].

The characteristic direction of site 93 ($D = -102.7°$, $I = 43.7°$) is far removed from the other directions of the Matoushan Formation after correction for tilt. Its direction before tilt correction ($D = 45.8°$, $I = 76.6°$) is remote by more than 30° from the tilt-corrected mean direction of this formation and the present axial dipole field. The magnetization of the red beds at site 93 may reflect either the geomagnetic excursion before folding or secular variation after folding. We therefore exclude its paleomagnetic data from further consideration.

Puchanghe Formation (Lower Cretaceous) Thirteen sites have characteristic directions. These are systematically distributed along a small circle from a downward and westerly direction to an upward and easterly direction before tilt correction (Fig. 4). The dispersed directions converge to three groups after tilt correction. The first group of eight sites has a northeasterly direction with positive inclination, and the second group (site 82, 84 and 88) shows a southwesterly direction with negative inclination. The paleomagnetic directions of the two groups are nearly antipodal, indicating that they are normal (group of eight sites) and reversed (site 82, 84 and 88) polarity directions.

The remaining two sites (site 83 and 86) form the third group, whose direction is northwest with positive inclination. These directions are derived from the sequential localities of site 82 to site 86. Sites 83 and 86 are located just above the strata of sites 82 and 84 with the reversed polarity of magnetization. Samples at site 85 reveal the coexistence of reversed and normal polarity directions. The red beds at sites 83 and 86 probably record either both polarities of magnetization within a single sample or the geomagnetic polarity reversal. Paleomagnetic directions of the third group were excluded from further consideration.

The tilt-corrected mean paleomagnetic direction, except for the two data of sites 83 and 86, is $D = 43.7°$, $I = 36.2°$, $\alpha_{95} = 17.1°$. Puchanghe passes the fold test with 99% confidence ($k_2/k_1 = 8.1/2.6 = 3.1$).

4. Discussion

4.1. *Reliability of the paleomagnetic data*

The most striking result from this work is a large clockwise deflection in the declination ($D > 44°$) of the paleomagnetic directions in the Chuxiong area: both Upper Cretaceous Matoushan and Lower Cretaceous Puchanghe Formations revealed easterly paleomagnetic directions. The re-

liability of the easterly directions is ascertained through thermal demagnetization, the presence of normal and reversed polarities, and the fold test. The easterly direction is not due to internal deformation of the rocks because there is no evidence of deformation or metamorphism in the red beds at the outcrops of the sampling sites and no original alignments (linear or planar elements) are observed under the microscope. Measurements of the anisotropy of magnetic susceptibility support this:

(1) The susceptibility ellipsoid of the samples is rather triaxial in shape, with prolateness $(K_{max} - K_{int})/(K_{int} - K_{min})$ [35] varying between 0.13 and 5.5 and its an arithmetical mean of 1.1 ± 1.4.

(2) The maximum axes of the susceptibility ellipsoid are distributed randomly in the sphere (Fig. 3, (4)). The paleomagnetic direction is not parallel to that of the maximum axis.

The directions of declinations are plotted for individual sampling sites in Fig. 5. Although the declination varies between $-2°$ and 116°, the easterly direction is observed in the entire area around Chuxiong City. The areal variation in declination is not due to internal block rotation within the Chuxiong area but is instead probably due to simple tilt correction for strata with large angles of bedding dip: it should be noted that the bedding attitude of fourteen sites is greater than 30°. North–south trending fold axes may, to varying degrees, undulate in the Chuxiong area, and insufficient accuracy in the tilt correction might induce the dispersion of declinations in the paleomagnetic directions on the equal-area projection in Fig. 4. The formation mean in the paleomagnetic directions is most likely to be a characteristic direction of the whole area around Chuxiong.

Our paleomagnetic study shows that almost all samples of red beds in Chuxiong have preserved characteristic remanent magnetization of pre-folding origin: thermal cleaning in particular eliminated a post-folding remanent magnetization at low temperature. Fieldwork suggests that the folding occurred in the Paleogene [28,36]. It is

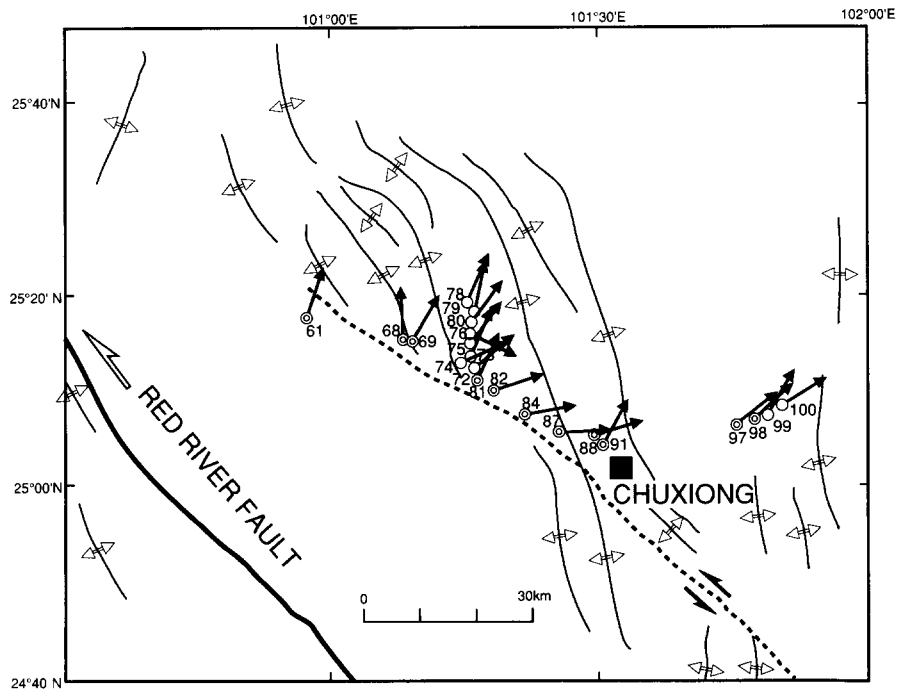


Fig. 5. Cretaceous paleomagnetic declinations at the sampling sites in the Chuxiong area. All the reversed directions have been inverted to normal directions. Single circle = sites of the Matoushan Formation (Upper Cretaceous); double circle = sites of the Puchanghe Formation (Lower Cretaceous) Number beside symbol is the name of the sampling site Bold line indicates the Red River fault and dashed line indicates a minor fault through the Chuxiong area. Tertiary anticlines are also shown (redrawn from [28]).

TABLE 2

Observed and expected paleomagnetic directions at Chuxiong

Area	Observed direction at Chuxiong					VGP			Ref.
	N	D_{obs}	I_{obs}	α_{95}	ΔD_{obs}	Lat.	Long.	α_{95}	
Western Yunnan Chuxiong (25.0°N, 101.5°E)	21	44.6°	41.3°	10.7°	14.3°	49.2°N	178.0°E	11 4°	
Area	Expected direction at Chuxiong					VGP			Ref.
	N	D_{exp}	I_{exp}	α_{95}	ΔD_{exp}	Lat.	Long.	α_{95}	
Whole China block	12	13.9°	39.6°	–	–	77.0°N	200.0°E	3.2°	[38]
Eastern part of the Yangtze block	5	19.1°	38.3°	7.8°	10.0°	72.1°N	198.6°E	7.2°	[39,40,41,42,43]
Western Sichuan	3	0.3°	26.2°	7.0°	7.8°	78.8°N	280.0°E	5.1°	[19,40,44]
Qiangtan Terrane (Markam)	3	46.7°	49.7°	9.6°	14.9°	48.6°N	172 9°E	12.6°	[19,21]
Lhasa Terrane	4	−11.6°	15.3°	14.3°	14.8°	69.5°N	316.6°E	13 3°	[15,17,19]

reasonable to conclude that the characteristic remanent magnetization of the Cretaceous formations of Chuxiong was acquired not much later than the time of deposition of the rocks. It seems likely that the easterly paleomagnetic direction is a record of the Cretaceous geomagnetic field, because major magnetic polarity in the Chuxiong area is normal, which is to be expected for Cretaceous time [37].

The mean paleomagnetic direction of the Matoushan Formation is consistent with that of the Puchanghe Formation; the difference in direction is only 6.7°. Good agreement of the paleomagnetic directions of the Upper and Lower Cretaceous formations is expected from the apparent polar wander path of the Eurasian continent, because Cretaceous time corresponds to the period of standstill in the wandering of paleomagnetic poles [38]. We therefore calculate a mean direction of the paleomagnetic directions of 21 sites in the two formations ($D = 44.6°$, $I = 41.3°$, $\alpha_{95} = 10.7°$), and recognize it as a characteristic

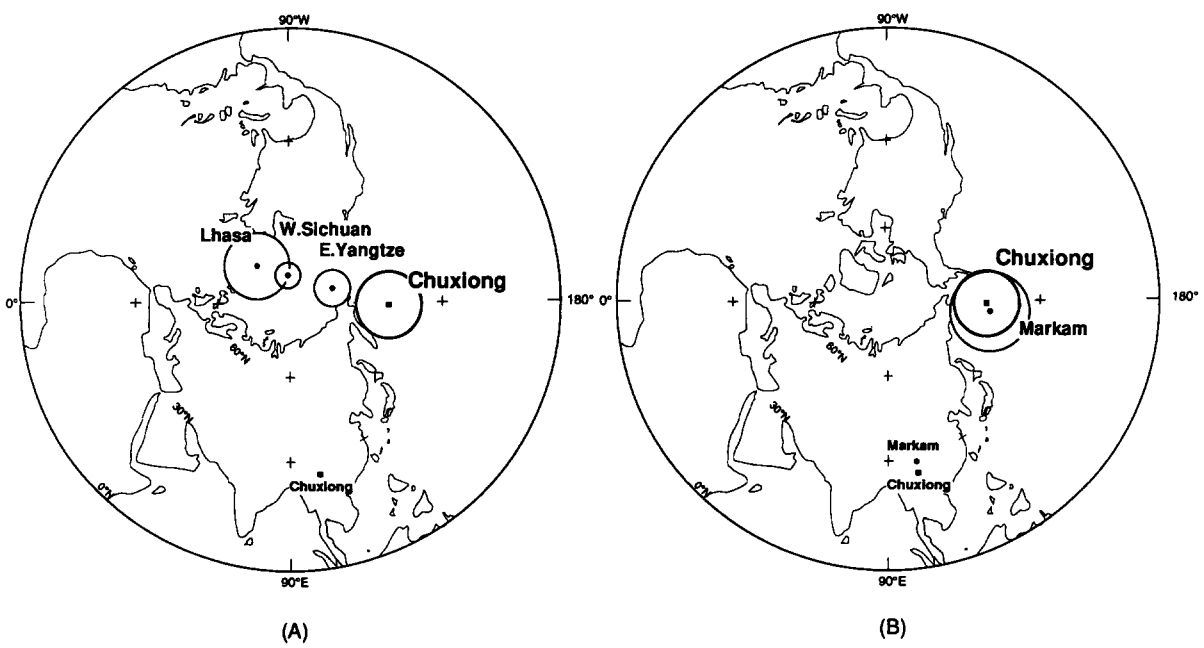


Fig. 6. Cretaceous paleomagnetic poles (with associated circles of 95% confidence) of the Chuxiong area and some other parts of Asia. (A) Lhasa Terrane, western Sichuan, eastern part of Yangtze. (B) Markam area of the Qiangtang Terrane. Pole and location of Chuxiong are shown by a square. Polar stereographic projection.

paleomagnetic direction of the Cretaceous in the Chuxiong area. These 21 sites yield a positive fold test which is significant at the 99% probability level.

4.2. *Clockwise rotation of the Chuxiong area*

The mean direction converted to virtual geomagnetic pole (VGP) for the Cretaceous of the Chuxiong area gives a latitude of 49.2°N, a longitude of 178.0°E and $\alpha_{95} = 11.4°$ (Table 2, Fig. 6). We compare our results with other Cretaceous poles from the Yangtze block and the Lhasa block in eastern Tibet (Fig. 6). We divide the Yangtze block into two blocks (the eastern part of the Yangtze block and western Sichuan) and calculate their VGPs. The VGP from the eastern part of the Yangtze block is located at 72°N, 199°E, $\alpha_{95} = 7°$ [39–43], whereas the VGP of western Sichuan (Ya'an and Huili; see Fig. 1) is at 79°N, 280°E, $\alpha_{95} = 5°$ [19,40,44]. The Cretaceous pole from the Lhasa block (70°N, 317°E, $\alpha_{95} = 14°$) [15,17,19] is located close to the pole for western Sichuan. The pole from the Chuxiong area is situated further east of all the poles from the Yangtze block and Lhasa block, clearly indicating that the Chuxiong area experienced large

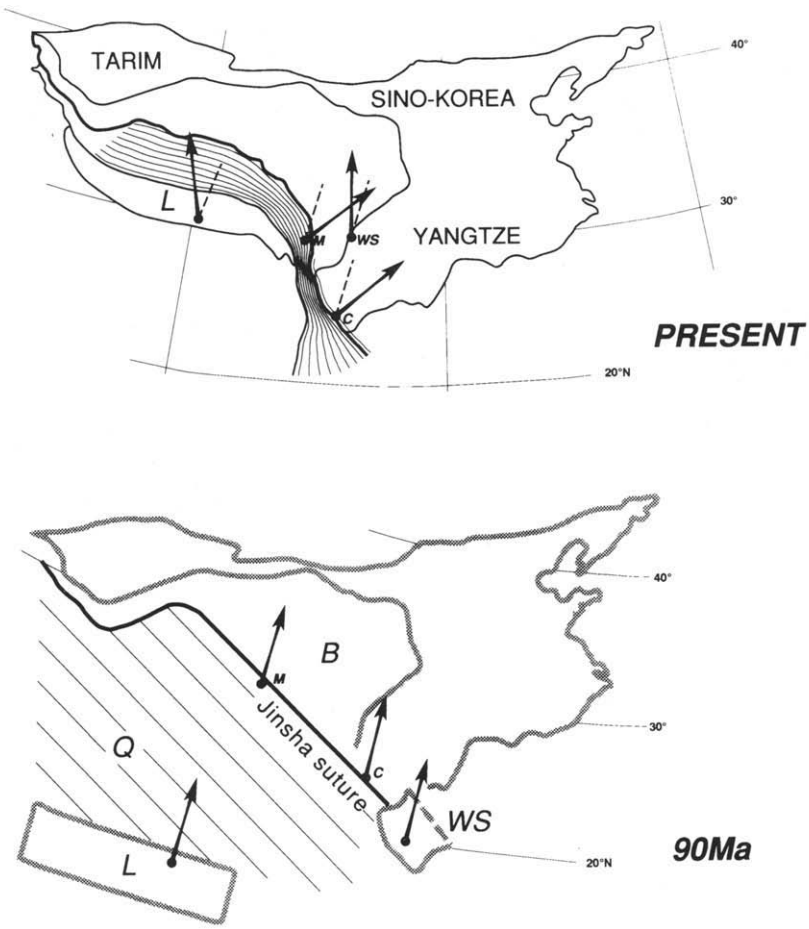


Fig. 7. Arrangement of tectonic blocks at the present and at 90 Ma (Cretaceous). *Present*. The solid arrow marks the observed declination and the dashed line the expected Cretaceous declination. *90 Ma* The three blocks of Lhasa (*L*), Qiangtang (*Q*) and western Sichuan (*WS*) are superimposed on the China block with the paleolatitudes of the Cretaceous. The strike of the Jinsha suture is assumed to have been WNW–ESE in this period. Chuxiong and Markam are placed on opposite sides of the Jinsha suture (dots). Chuxiong (*C*) and Markam (*M*) are arranged at 24°N and 28°N in the Cretaceous coordinate system. The paleolatitude of China is calculated from the pole at 90 Ma (44°N, 200°E) of Besse and Courtillot [38] The paleomagnetic direction lies along the Cretaceous paleomeridian *B* = Bayan Kala Terrane

clockwise rotational motion with respect to the three blocks since Cretaceous time.

The amount of rotation of the Chuxiong area may be evaluated from comparison of the observed declination value and the expected declination values calculated from the VGPs of the three blocks (Table 2). The amount of rotation R and its uncertainty ΔR are given by:

$$R = D_{obs} - D_{exp};$$

$$\Delta R = \left[(\Delta D_{obs})^2 + (\Delta D_{exp})^2 \right]^{1/2}$$

where D_{obs} and D_{exp} are observed and expected declination, and ΔD_{obs} and ΔD_{exp} are their uncertainty at 95% confidence ($\Delta D = \sin^{-1}$ (sin α_{95}/cos I), with α_{95} being the radius of 95% confidence circle and I the inclination). The Chuxiong is rotated clockwise through $26° \pm 17°$ with respect to the eastern part of the Yangtze block. The amount of the rotation of Chuxiong exceeds 44° with respect to western Sichuan ($44° \pm 16°$) and the Lhasa block ($56° \pm 21°$). Large relative rotation occurred with the neighboring regions rather than with the eastern part of the Yangtze.

Rotation of the Chuxiong area must be a feature of the tectonics for a fairly large area, including the eastern Qiangtang Terrane. A large clockwise deflection in declination ($D = 48°$) has been reported from the Cretaceous rocks of the Markam area (29.7°N, 98.6°E) in the Three Rivers region on the Qiangtang Terrane (Fig. 7) [19,21]. Although 500 km lie between the Markam area and Chuxiong, the Cretaceous pole from Markam (49°N, 173°E, $\alpha_{95} = 14°$) is located close to the pole from Chuxiong (49°N, 178°E) (see Fig. 6). The good agreement of the poles indicates that little relative rotation or translation occurred between the Chuxiong area and the Markam area. This implies that the two areas experienced similar tectonic movement within the Asian continent.

The coherent rotation of both the Chuxiong and Markam areas receives support from the shape of the Jinsha suture–Red River fault, which arches to the northeast, forming a convex arc (Fig. 7). The Jinsha suture runs along N70°W on the Tibetan Plateau and its strike changes to NNW–SSE (N10°W in Markam and N35°W in Chuxiong) in eastern Tibet and Yunnan. The magnitude of the bend (35°–60°) in both the suture and the fault is compatible with the paleomagnetic rotation ($\sim 56°$) of the Markam and Chuxiong areas with respect to the Lhasa block in the Tibetan Plateau. The Markam and Chuxiong areas were coherently rotated as essentially a single tectonic block together with the Jinsha suture and Red River fault.

The timing of the rotation may be estimated using the paleomagnetic data and morphological features: the tectonic rotation occurred prior to the folding of central Yunnan. Paleomagnetic data show that the Chuxiong area experienced large clockwise rotation, whereas little rotation occurred in the interior of the Yangtze block 250 km from the Red River fault. Kent et al. [40] reported northerly declination ($D = -2.4°$) in the Huili area (26.5°N, 102.3°E), which is about 190 km north of Chuxiong. This indicates that the Chuxiong (25.0°N, 101.5°E) and Huili (26.5°N, 102.3°E) areas have undergone relative rotation of more than 45°. The relative rotational motion, however, is not recorded in the orientation of the fold axes (see Fig. 1): the axes of the folds trend north–south in the mountainous area between Chuxiong and Huili [28,45]. This implies that the clockwise rotational event preceded the folding in central Yunnan. We conclude that the rotation occurred during or before the Paleogene.

The large clockwise rotation of the Chuxiong area is not ascribed to local deformation associated with strike slip. Sampling sites are distributed along the minor strike-slip fault in the Chuxiong area, which is located on the northeastern side of the Red River fault. No model of local deformation accounts for any of the characteristics of the large rotational motion—neither the direction of rotational nor the amount or timing of rotation: First, clockwise rotation is generally expected for crustal blocks that move along a strike-slip fault with right-lateral movement [46]. A crustal block bounded by sets of parallel faults with right-lateral slip also experiences clockwise rotation [47–50]. Both the minor fault and the Red River fault are, however, expected to have been left-lateral in the Tertiary [28]. Second, there is no correlation between the amount of paleomagnetic rotation and distance from the minor fault (Fig. 5), indicating that the strike-slip induced little drag on the crustal blocks along the

minor fault. Third, on the basis of U-Pb radiometric ages the time of occurrence of the large left-lateral strike-slip of the Red River fault is estimated to be about 23 Ma [45]. Some monor right-lateral strike-slip faulting for the Red River fault is postulated for the Quaternary [50]. The Paleogene clockwise rotation of the Chuxiong area is not accounted for by strike-slip of the Red River fault in the Oligo-Miocene or Quaternary. Clockwise rotation is, therefore, ascribed to the regional rotation rather than to local deformation.

4.3. Continental deformation of South China

Cretaceous paleopositions of the Chuxiong area and its vicinity are arranged with respect to the China block on the basis of both declinations and inclinations (Fig. 7). We assume that the China block consists of the Tarim, the eastern part of the Yangtze, and the Sino-Korea Craton [23,51,52], and that its Cretaceous pole position was at 77°N, 200°E [38].

The Qiangtang and Chuxiong areas have been arranged in approximately their present positions with respect to the China block, because the paleomagnetic data from these areas show fairly high latitudes in the Cretaceous: the paleolatitudes of the Chuxiong and Markam areas are estimated to be 24°N (±12°) and 31°N (±13°), respectively. We place Chuxiong and Markam at 24°N and 28°N in the Cretaceous coordinate system. Extrapolating the strike at the eastern end of the Red River fault into the Tibetan area, we presume that the Jinsha suture and Red River fault ran WNW–ESE in the Cretaceous. Because the Chuxiong and Markam areas stand on opposite sides of the Jinsha suture, the positions of the two areas are consequently pulled apart without latitudinal change in the Cretaceous arrangement. Our reconstruction places the Lhasa and western Sichuan blocks further to the south than the present positions—at about 14°N and 19°N respectively.

This arrangement leads us to propose the following tectonic evolution: The Lhasa and western Sichuan blocks moved northward after the beginning of the late Cretaceous. While the large translation of the Lhasa block shortened the width of the central part of the Qiangtang Terrane, the eastern Qiangtang Terrane around the Markam moved east toward the western Sichuan block. Seeing as the Chuxiong area has occupied its present position since the Cretaceous, the eastward movement of the eastern Qiangtang Terrane caused a block consisting of the Markam and Chuxiong areas to rotate clockwise with little relative latitudinal translation. The block rotation produced the bent form of the Jinsha suture–Red River fault in the Three Rivers region. Consequently, the deformation which originated in the Lhasa and Qiangtang Terranes propagated into the interior of the Yangtze block across the Red River fault.

The northward translation of the Lhasa block is ascribed to the indentation of the Indian continent [17]. The timing of the clockwise rotation (ca. Paleogene) in the Chuxiong area is consistent with the period of the collision of India (~45 Ma) [2,3]. The rotation of the Qiangtang and Chuxiong areas is, therefore, ascribed to the collision between the Asian and Indian continents.

In all tectonic models studied using laboratory and numerical experiments for the collisional deformation of the Asian continent [e.g. 6–9,53,54] clockwise rotation of the terranes and faults is expected within the Asian continent near the eastern corner of the colliding Indian continent. Our paleomagnetic data strongly support the theoretical prediction that rotation is detectable within 1000 km of the indenter [9], because Chuxiong is separated by about 550 km from the corner of rigid India at 28°N, 96°E. In further elaborations of these tectonic models, deformation processes are divided into two categories—deformation of a rheological continuum [6,9,53] and lateral crust extrusion [8,10]. The following three observations show that the rotational motion in South China is achieved by the former model:

First, the deformation extended 100 km inland of the Yangtze block across the Red River fault, and the margin of the Yangtze block, with its width of 100 km, behaved as a non-plate-like material in a continuum during the collision events.

Second, the relative rotation of more than 45° is accommodated in continuous deformation within crustal material between Chuxiong and Huili. The rotated Chuxiong area is not bounded by any strike-slip faults in the north.

Third, the Red River fault does not play an important role in the rotation of the Chuxiong area.

We conclude that the rotation of the Markam and Chuxiong areas is a manifestation of deformation phenomena of continuum material within the Asian continent due to collision with India.

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